Belief Geometry on the Random Hierarchy Model

A small transformer partially encodes — and causally uses — the exact tree posterior, probed against ground truth

simplex-rhm-belief · run of 29 June 2026 · all numbers from the recorded pipeline output

Abstract. We reproduce, in miniature, the Simplex belief-geometry result on a hierarchical generative process. We train a 2-layer GPT-2 decoder (≈399k parameters) by next-token prediction on samples from a Random Hierarchy Model (RHM; arity $s = 2$, depth $L = 3$, vocabulary $v = 8$), a grammar small enough (1024 equiprobable trees) to enumerate exactly. We compute the **exact** Bayesian posterior over the tree's hidden latents by sum-product belief propagation, verified against brute-force enumeration to $< 10^{-6}$. The trained model reaches a test cross-entropy of 0.89 nats, closing ≈88% of the gap between the uniform baseline (2.08) and the Bayes-optimal floor (0.73) — it has effectively learned the posterior predictor. A single global linear probe partially recovers this posterior from the residual stream: held-out $R^2 = 0.38$ for the root class versus ≈0 for a shuffled control — a real but noisy affine image, not an exact reconstruction — with decodability accumulating monotonically across depth ($-0.00 \to 0.15 \to 0.38$) and the belief readout **blooming** outward from the prior toward simplex vertices as context accumulates. A latent-level sweep shows the residual encodes the **whole** hierarchy, most strongly the locally-predictive deep latents (R^2 up to 0.66) and least strongly the coarse global root. Causal steering on the layer-1 residual confirms the belief is **used**, not merely decodable: steering toward a wrong latent collapses the true next token's log-probability ($-0.73 \to -8.5$). We pre-registered our predictions before seeing results; three of four held, and the one miss (root $R^2 < 0.5$) is informative. Finally, a 30-run sweep over 10 random grammars × 3 replicate seeds shows all four findings — the inverted level gradient, blooming, layer accumulation, and causal steering — replicate across the RHM family with low replicate variance, and that the published `grammar = 0` numbers sit inside the family distribution.

1. Introduction

Natural data is hierarchical: characters compose into words, words into phrases, phrases into meaning. The Random Hierarchy Model (RHM) of Cagnetta et al. abstracts this into a clean synthetic grammar — a fixed tree in which each high-level symbol expands, via random production rules, into a fixed-length string of lower-level symbols, down to observed leaves. Predicting the next leaf optimally requires inferring the distribution over the **hidden** latent symbols that generated the observed prefix. The optimal next-token predictor **is** the Bayesian belief state over those latents.

This makes the RHM an ideal testbed for **belief geometry**: the hypothesis, developed in the transformer-interpretability literature, that a network trained by next-token prediction comes to represent, in its residual stream, the posterior distribution over the data-generating process's hidden state — and that this distribution is laid out as a linear (affine) image of the probability simplex. If true, the belief should be (i) linearly decodable, (ii) geometrically organized as a simplex that sharpens with context, (iii) built up additively across layers, and (iv) causally relevant to the model's output.

The RHM is uniquely suited to test all four claims **exactly**. Unlike natural language, its hidden state has a precise definition and its posterior is computable in closed form by belief propagation on the tree. With small parameters the entire grammar is enumerable, so the probe target is ground truth — not an estimate. This paper trains a tiny transformer on such a grammar and asks whether the residual stream carries the exact posterior, where in the network it lives, what geometry it traces, and whether the model relies on it.

2. The Random Hierarchy Model and exact beliefs

Grammar. We use the defaults of the spec: arity $s = 2$, depth $L = 3$, so each string has $d = s^L = 8$ leaves; per-level vocabulary $v = 8$; and $m = 2$ production rules per parent symbol, chosen uniformly. Rules are drawn once and held fixed, and constrained to be **unambiguous** (the spec's quick-failure check confirms no two parents share a production). A sample is generated top-down: pick a root class uniformly, recursively expand each symbol by a uniformly chosen rule, and emit the leaf string. With these parameters the grammar admits exactly $v \cdot m^{d-1} = 8 \cdot 2^7 = 1024$ equiprobable distinct trees — small enough to enumerate completely.

The exact frozen grammar used in every result is shown below. Symbols are written as `S1`, …, `S8` for readability (the saved arrays use zero-based ids); at each level the parent takes one of the two listed child pairs uniformly.

Parent	Top expansion	Middle expansion	Bottom expansion
`S1`	`[S3, S1]` or `[S4, S8]`	`[S2, S1]` or `[S1, S5]`	`[S4, S1]` or `[S5, S3]`
`S2`	`[S4, S3]` or `[S1, S3]`	`[S6, S3]` or `[S3, S1]`	`[S6, S7]` or `[S8, S8]`
`S3`	`[S6, S7]` or `[S5, S6]`	`[S5, S2]` or `[S1, S1]`	`[S3, S4]` or `[S7, S8]`
`S4`	`[S6, S8]` or `[S7, S7]`	`[S6, S5]` or `[S8, S6]`	`[S6, S5]` or `[S4, S7]`
`S5`	`[S6, S2]` or `[S1, S5]`	`[S4, S3]` or `[S1, S8]`	`[S1, S5]` or `[S6, S4]`
`S6`	`[S8, S6]` or `[S2, S7]`	`[S7, S6]` or `[S4, S5]`	`[S8, S4]` or `[S1, S8]`
`S7`	`[S4, S7]` or `[S1, S1]`	`[S3, S8]` or `[S4, S1]`	`[S4, S6]` or `[S3, S2]`
`S8`	`[S2, S2]` or `[S5, S7]`	`[S1, S2]` or `[S8, S3]`	`[S5, S7]` or `[S3, S6]`

Table 1: Frozen level-specific grammar sampled once before training. Top, middle, and bottom columns correspond to `rules_0`, `rules_1`, and `rules_2` in `artifacts/grammar.npz`.

Exact posterior. Given a leaf prefix of length k, the posterior over any hidden latent (including the root class) is computed by sum-product belief propagation on the tree: upward messages from observed leaves combine through the production rules to give the marginal over each latent. Because the grammar is fully enumerable, we verified the belief-propagation posterior against brute-force enumeration of all consistent trees; the two agree to $< 10^{-6}$ across 10 passing tests, alongside checks that sample length equals s^L and that unambiguity holds. The recorded self-test (a representative string) shows the root posterior sharpening with context — entropy 1.89 nats at $k = 1$ falling to 0.00 (a single consistent root) by $k = 3$ — which previews the blooming geometry below.

3. Model and training

We train a HuggingFace `GPT2LMHeadModel` with 2 layers, `n_embd` = 128, 4 heads, and all dropout disabled, totalling 398,848 parameters. There is no tokenizer: RHM leaf symbols are integers fed directly as `input_ids`, with `vocab_size` = 8 and `n_positions` = 8. We train by next-token prediction on 922 training strings (102 held out) for 4000 steps on Apple MPS.

The Bayes-optimal floor. The value of an enumerable grammar is that we know the best achievable loss. Averaging the exact per-position posterior entropy over the seven predicted positions gives the Bayes-optimal mean next-token cross-entropy: 0.725 nats. The uniform baseline is $\ln 8 = 2.079$. The trained model reaches a final held-out cross-entropy of **0.891 nats** (Figure 1), closing $\frac{2.079-0.891}{2.079-0.725} \approx 88\%$ of the gap. The model has, to a good approximation, learned the true posterior predictor — which is the precondition for asking whether it represents the posterior internally.

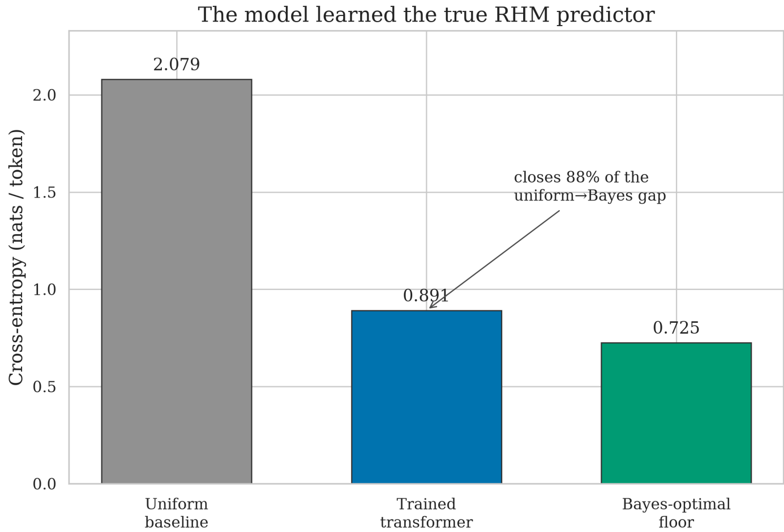

Figure 1: The transformer learned the true RHM predictor. Test cross-entropy (0.891 nats) sits far below the uniform baseline ($\ln 8 = 2.079$) and close to the Bayes-optimal floor (0.725), closing ≈88% of the uniform→Bayes gap.

4. Pre-registered prediction

Following the project's honor-code rule, we committed our predictions to git **before** training any model or computing any probe (`PREREGISTRATION.md`). In brief, we predicted: (1) the root posterior is **linearly decodable** from the residual stream, with $R^2 > 0.5$ at the final layer and a shuffled baseline near 0; (2) the belief **blooms** — early positions cluster near the prior, late positions spread toward simplex vertices, with posterior entropy falling and activation radius growing monotonically with context k; (3) decodability **accumulates additively across depth**, lowest at the embedding and highest after the last layer; and (4) the representation is **causally used**, so steering the residual along a belief direction shifts the output toward the corresponding latent's leaves. We also registered alternative outcomes (degenerate collapse, non-linear-only encoding, MAP-only encoding, flat-with-depth) as falsification handles. We report against these predictions in Section 6.

5. Results

5.1. A single linear probe partially decodes the posterior

We fit one global least-squares affine map from the 128-d residual stream to the 8-class exact root posterior, on a probe set of $N = 400$ held-out examples, and score it by R^2 on held-out data. The final-layer probe reaches $R^2 = 0.38$, while the same probe fit to **shuffled** labels scores ≈ 0 (-0.085). So the posterior is genuinely present and linearly accessible above chance — but $R^2 = 0.38$ means the probe explains only about a third of the variance: this is a **partial**, noisy recovery, not an exact reconstruction. Projected into a belief-space PCA basis (Figure 2), the probe's predicted posteriors occupy the same structured region as the exact posteriors and reproduce the same position gradient, but as a diffuse cloud rather than the discrete point set of the ground truth. (The ground-truth panel looks sharper partly because the exact posterior takes few distinct values, so identical points overplot, whereas every probe prediction differs slightly.) The affine correspondence is real, but for the global root it is weak; it strengthens markedly for deeper latents (next subsection) and at later layers.

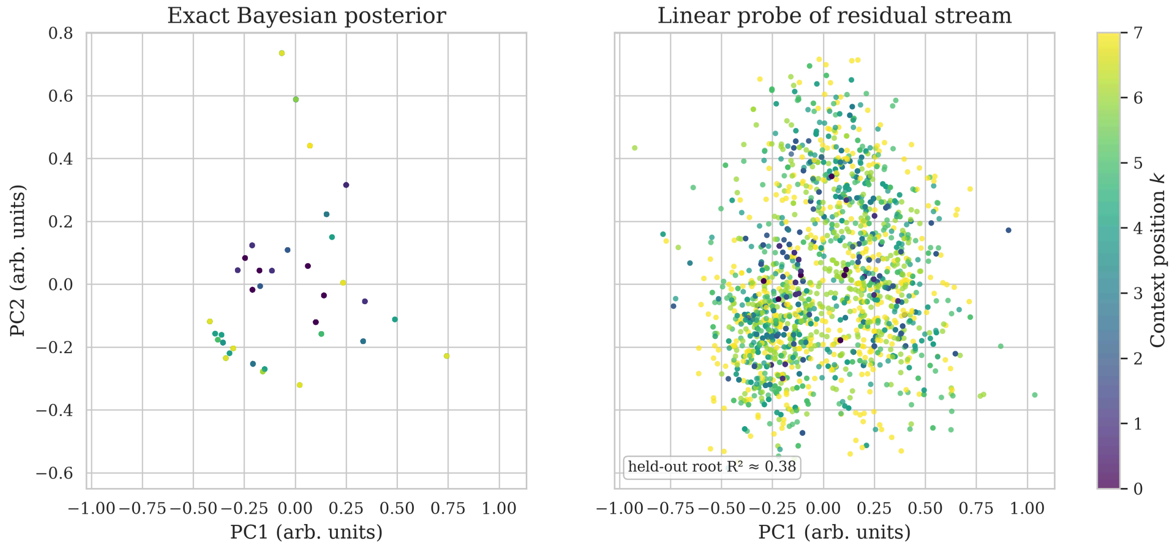

Figure 2: The residual stream is a **partial** affine image of the exact belief simplex. **Left:** PCA(2) of the exact root posteriors (few distinct values, hence sharp overplotted dots); small-k points sit near the prior, large-k points spread toward vertices. **Right:** the linear probe's predictions in the same basis, colored by context position — the same region and position gradient, but a diffuse cloud: held-out root $R^2 \approx 0.38$, an imperfect recovery, not an exact one.

5.2. Decodability accumulates across depth and context

The residual stream is a running sum of layer contributions, so we ask where the belief is built. Probe R^2 for the root posterior rises monotonically with depth: -0.00 at the embedding (layer 0), 0.15 after layer 1, 0.38 after layer 2, while the shuffled control stays at ≈ 0 throughout (Figure 3, left). Each layer adds belief-relevant signal. Resolving by context position (Figure 3, right), early positions are decodable even at shallow layers, and the final layer pushes near-perfect decodability ($R^2 = 1.0$) across positions 0–3. A few mid-layer cells are strongly negative (the probe underperforms the mean predictor on those positions); the heatmap colors are clipped for legibility, but the cell labels show the raw values.

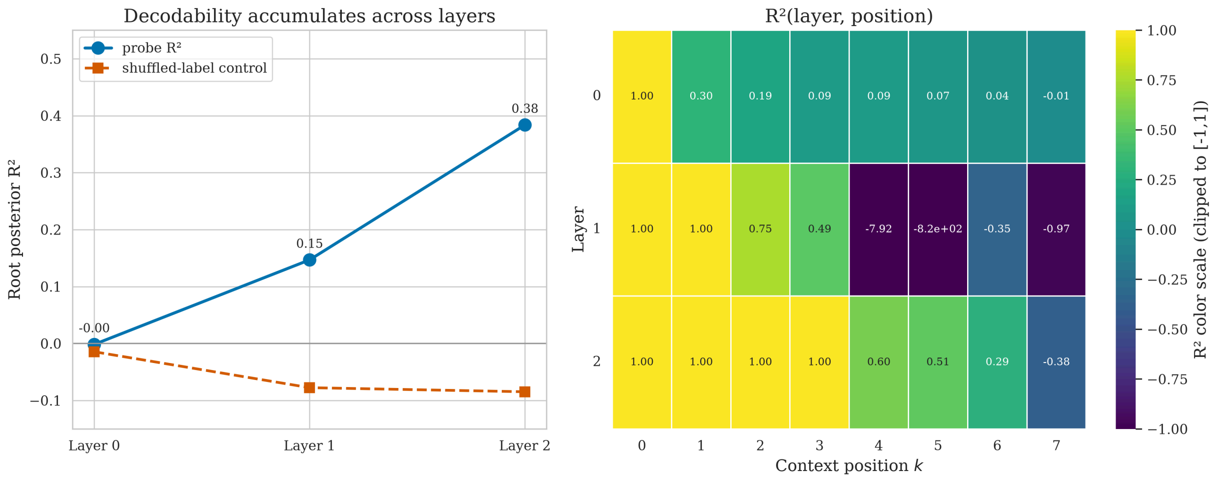

Figure 3: Belief decodability accumulates across depth and context. **Left:** root-posterior R^2 rises layer by layer ($-0.00 \to 0.15 \to 0.38$) while a shuffled control stays at ≈0. **Right:** R^2(layer, position) heatmap; the final layer reaches $R^2 = 1.0$ on the earliest positions. Colors are clipped to $[-1, 1]$, while labels show raw values.

5.3. The whole latent hierarchy is encoded — local latents most strongly

The root is only one of the tree's hidden latents. Probing the exact posterior over latents at **every** level reveals a clear gradient with some node-level variation (Figure 4): root (L_0) $R^2 = 0.38$; the two level-1 mid latents 0.51 and 0.59; and the four deepest level-2 latents 0.61, 0.66, 0.50, and 0.66. The residual encodes the entire hierarchy, but the **local, near-leaf** latents — those most directly predictive of the next token

— are usually read off more cleanly than the coarse global root. This is our most informative finding: the spec's primary target (the root) is the **hardest** latent to decode, because it is the most abstract and the least locally predictive.

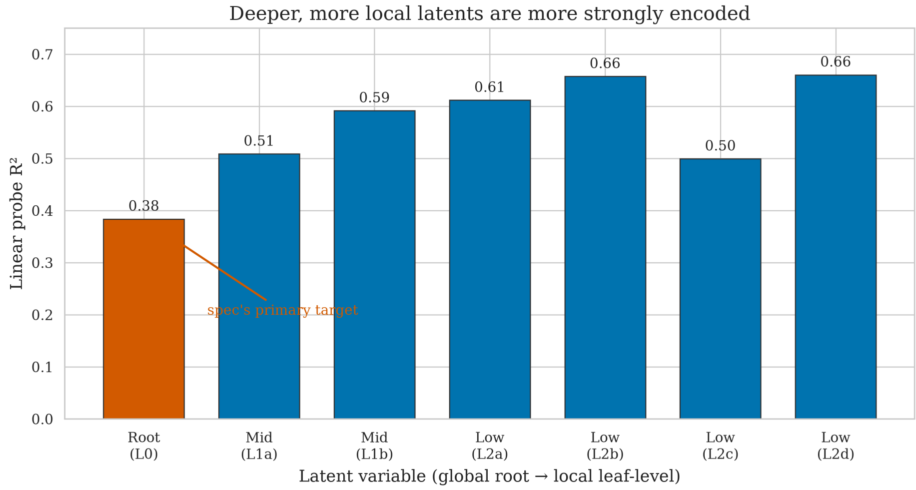

Figure 4: The residual encodes the whole latent hierarchy, deeper/local latents most strongly. Probe R^2 is lowest for the root ($L_0 = 0.38$, the spec's primary target), higher for level-1 $(0.51, 0.59)$, and generally higher for level-2 $(0.61, 0.66, 0.50, 0.66)$.

5.4. The belief blooms with context

The headline geometry (Figure 5): projecting the linear readout into a belief-space PCA(2) basis, the points form a tight central cluster near the uniform prior when little context is observed and expand outward toward the simplex vertices as context accumulates. The mean radius about the prior anchor grows monotonically from 0.17 at $k = 0$ to ≈0.39 at $k = 7$, tracking the true posterior's own growth (0.17 → 0.47); equivalently, the exact posterior entropy falls monotonically with position (1.84 nats at $k = 0$ to 0.00 at $k = 7$). Colored by ground-truth root class, the cloud resolves into separated petals, one per inferred root. Raw residual PCA, dominated by token and position nuisance variance, does **not** bloom — it is the **belief content** of the residual that does.

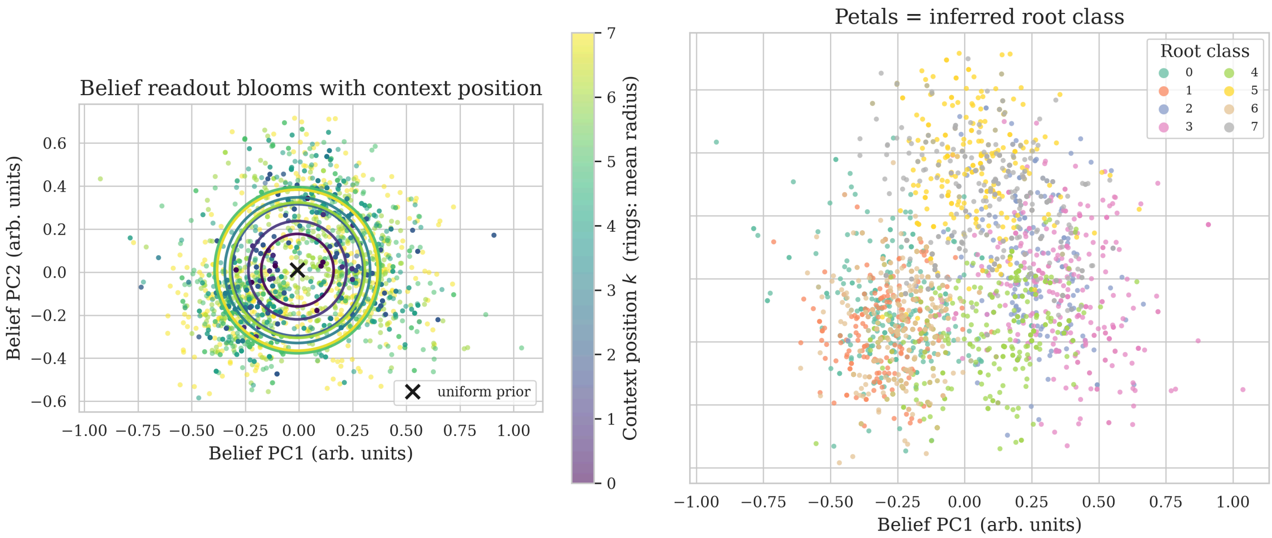

Figure 5: Headline result: the belief read out of the final-layer residual blooms with context. **Left:** points colored by context position k with rings marking each position's mean radius about the prior (×); the cloud expands outward as context grows (mean radius 0.17 → 0.39, tracking the true posterior 0.17 → 0.47). **Right:** the same cloud by ground-truth root class resolves into separated petals.

5.5. Causal steering: the belief is used, not just decodable

A representation can be decodable yet causally inert. To distinguish the two we apply a mean-difference patch to the layer-1 residual, pushing it by α times the belief direction toward a chosen latent, and measure the effect on the next-token distribution (Figure 6). Steering **toward the true** latent leaves predictions untouched (the model already holds that belief): the true token's log-probability stays at -0.73 for all α. Steering **toward a wrong** latent collapses the mean log-probability of the true next token from -0.73 to -8.5 as α grows, and re-routes probability mass onto the wrong latent's children (from 0.22 to 0.53). Intervening on the decoded belief axis changes behavior in the predicted direction — the belief state is causally used.

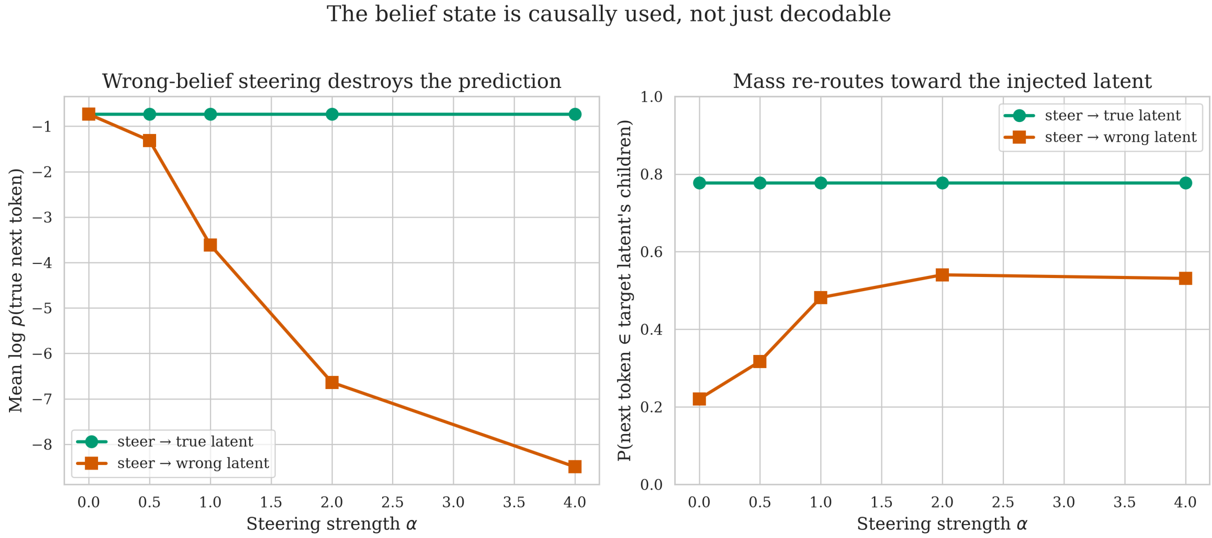

Figure 6: The belief state is causally used. Steering the layer-1 residual toward the **true** latent leaves predictions intact (green); steering toward a **wrong** latent collapses the true next token's log-probability (left, orange: $-0.73 \to -8.5$) and re-routes mass onto the wrong latent's children (right, orange: $0.22 \to 0.53$).

6. Pre-registration scorecard

We grade each pre-registered prediction honestly against the outcome.

Prediction	**Outcome**	**Verdict**
Linear decodability, root $R^2 > 0.5$, shuffled ≈0	Root $R^2 = 0.38$ (shuffled ≈ 0). Decodable and well above baseline, but below the 0.5 threshold for the root; **mid/deep latents do exceed 0.5** (up to 0.66).	Partial
Blooming with context (radius ↑, entropy ↓, monotone)	Mean radius $0.17 \to 0.39$ monotone; posterior entropy $1.84 \to 0.00$ monotone.	Hit
Monotone layer-wise accumulation of R^2	$-0.00 \to 0.15 \to 0.38$ across the three residual indices.	Hit
Causal usage via steering (ordered effect)	Steering→wrong: log p collapses $-0.73 \to -8.5$, monotone in α; target-child mass $0.22 \to 0.53$.	Hit

Three of four predictions held cleanly. The single miss is the root R^2, which came in at 0.38 rather than the predicted > 0.5. We take this as a genuine and informative negative: the prediction was correct in **form** (linear, above baseline, growing with depth and context) but our magnitude estimate for the **root specifically** was optimistic. The latent-level sweep — which we did not pre-specify — explains why: the root is the coarsest, least locally-predictive latent, and the threshold we predicted is in fact met by every deeper latent in the hierarchy. None of the pre-registered failure modes (degenerate collapse, non-linear-only encoding, MAP-only encoding, flat-with-depth) materialized; in particular MAP root-class accuracy is only 0.47 while full-simplex R^2 is positive and graded, ruling out a MAP-only representation.

7. Generalization across grammars

The results above come from a single grammar (`grammar = 0`) and a single training run. To test whether they are properties of the RHM **family** under our fixed constraints ($s = 2$, $L = 3$, $v = 8$, $m = 2$, uniform unambiguous rules) rather than artifacts of one rule draw, we sweep the **content of the rule table** across ten independently sampled grammars (`Grammar.random(seed = 0..9)`) with three replicate training seeds each — 30 runs at the pinned architecture ($n_{\text{layer}} = 2$, $n_{\text{embd}} = 128$, $n_{\text{head}} = 4$) and full 4000-step budget. A single master seed controls model initialisation, training-data sampling, probe sampling, and the probe split, so replicates measure end-to-end sampling variance. Each run writes a deterministic, self-contained directory (`results/sweep/g{NN}_L2_d128_h4_s{S}/`) holding its grammar, a `config.json` manifest (resolved config, git SHA, timestamp), and per-run metrics — recoverable for this paper independent of any experiment-tracker. Published `grammar = 0` is the first cell of the sweep, so its number is locatable in the distribution as a built-in consistency check.

The ten grammars are genuinely distinct. Our level-granularity metrics are unchanged if a grammar's symbols are renamed, so a sweep over mere relabelings of one grammar would prove nothing. It is not such a sweep. An exact isomorphism test on the rule tables (`grammar_iso.py`, checkpointed in `tests/`) finds all $\binom{10}{2} = 45$ pairs non-isomorphic — even allowing a global left↔right child swap — and correctly flags a grammar against a relabeled copy of itself. The between-grammar spread of root R^2 (SD 0.055) likewise exceeds the replicate noise (SD 0.040): the grammars differ in real inference difficulty, not just labels.

All grammars train to near-Bayes. Every run closes 0.82–0.93 of the uniform→Bayes loss gap (mean 0.89 ± 0.03), matching the pass-1 value (0.88); no run failed to train, so we report all 30 with no convergence filtering (Table 2). Loss-gap-closed is shown as a sanity covariate, not used to gate any run.

The inverted strength gradient replicates as a distribution (Figure 7). Pooling probe R^2 by tree level, the root (L_0) averages 0.35 ± 0.07 (range 0.24–0.48), well below the mid (L_1, mean 0.50) and leaf-parent (L_2, mean 0.49) latents. The root is the weakest-decoded level across the **whole family**, not just in the published draw: the coarse global latent is the hardest to read off, while the locally-predictive deeper latents are encoded more strongly. The pass-1 root $R^2 = 0.38$ sits comfortably inside the replicate band, confirming it as a typical rather than cherry-picked draw. The deepest level shows the widest spread (one grammar dips slightly negative), as expected — node-level rule structure matters most for the most local latents.

Blooming and belief-sharpening hold for every grammar (Figure 8). The exact posterior entropy falls monotonically with context position k in all ten grammars, and the belief readout's radius grows with context — the blooming geometry is a family-wide property, not a feature of one rule table.

Causal steering replicates with a stable effect size (Figure 9). Steering the layer-1 residual toward a wrong latent collapses the true next token's mean log-probability in every run, by 6.9 ± 0.5 nats from $\alpha = 0$ to $\alpha = 4$. The belief is causally used across the whole grammar family, with low run-to-run variance.

In short, all four pass-1 findings — inverted level gradient, blooming, layer-wise accumulation (recomputed in every run's per-layer probe), and causal steering — replicate across the grammar family and are stable across replicate sampling. The headline phenomenology is a property of the RHM under these constraints, not of one rule draw.

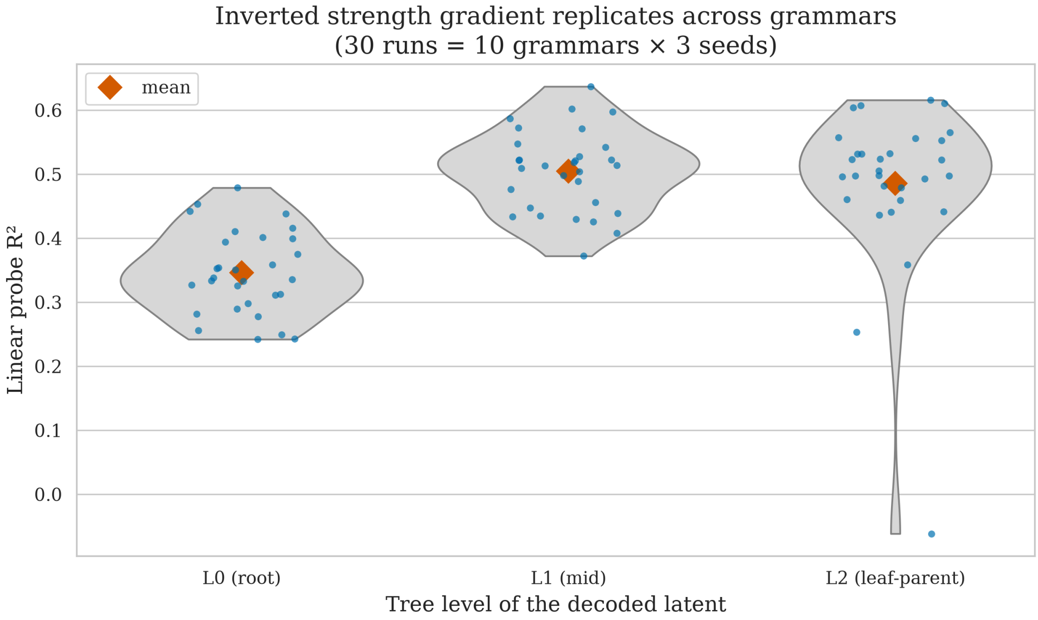

Figure 7: The inverted strength gradient replicates across the grammar family. Each point is one of 30 runs (10 grammars × 3 seeds); violins show the per-level distribution of probe R^2, diamonds the means. The root (L_0) is the weakest-decoded level (0.35 ± 0.07), below the mid (L_1) and leaf-parent (L_2) latents (≈ 0.49–0.50).

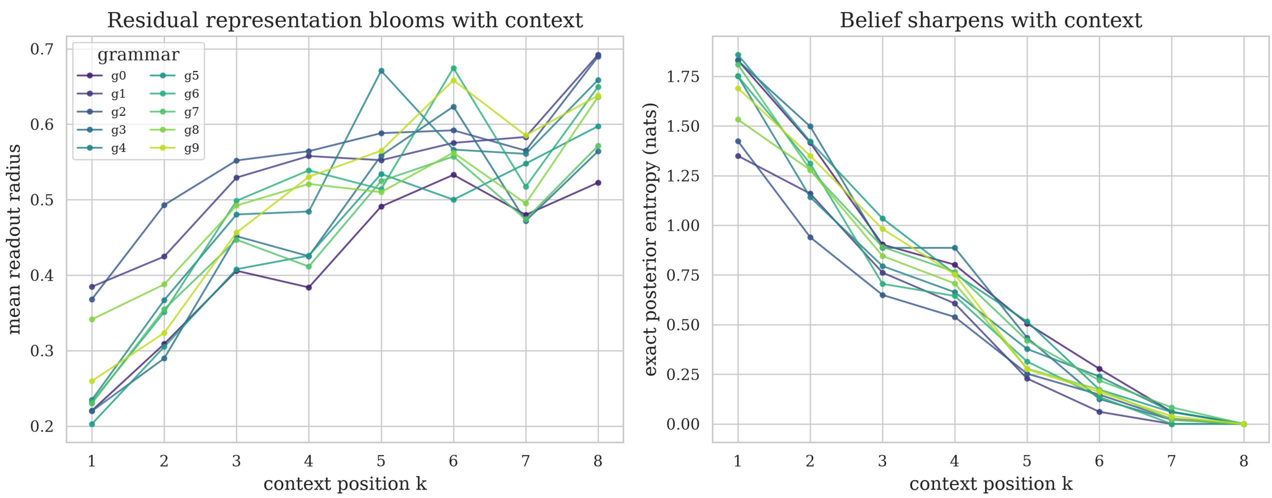

Figure 8: Blooming is family-wide. **Left:** the belief readout's mean radius grows with context position k (one line per grammar, averaged over seeds). **Right:** the exact posterior entropy falls monotonically with k for every grammar.

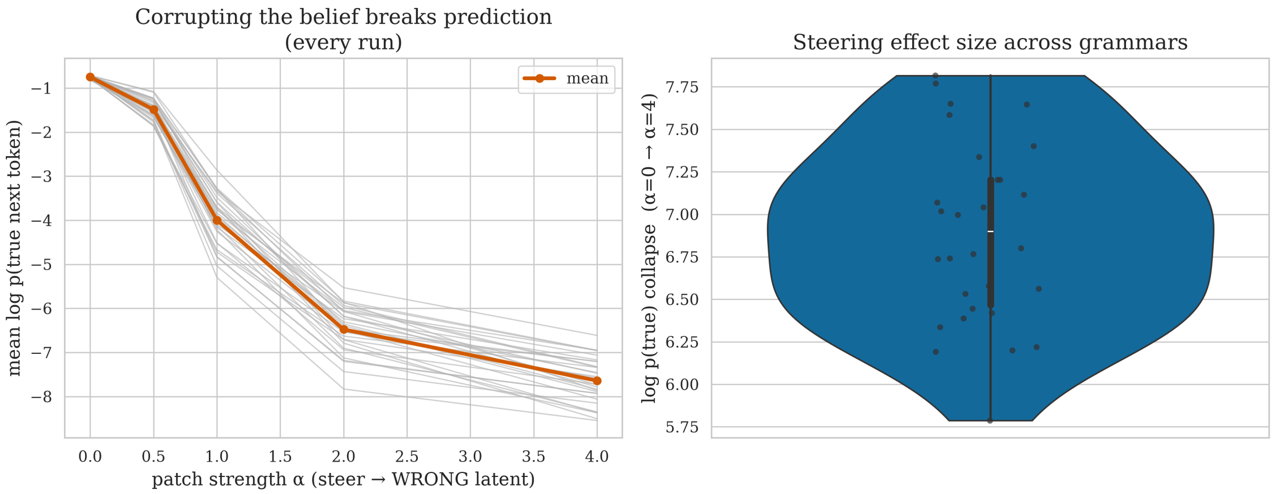

Figure 9: Causal steering replicates with a stable effect size. **Left:** steering the layer-1 residual toward a **wrong** latent collapses the true next token's log-probability in every run (grey lines), mean in orange. **Right:** the distribution of the collapse magnitude ($\alpha\{=\}0 \to \alpha\{=\}4$) across all 30 runs: 6.9 ± 0.5 nats.

8. Discussion and limitations

The picture is coherent: a tiny transformer trained to near-Bayes-optimal loss on a hierarchical grammar carries a **partial** linear image of the posterior over the grammar's hidden latents in its residual stream — fuzzy for the coarse root ($R^2 = 0.38$), sharper for the local latents (R^2 up to 0.66). That image is assembled additively across layers, blooms from the prior toward the vertices as evidence accumulates, spans the full latent hierarchy, and is causally relied upon at generation time. We probe against an **exactly known** belief state (computed by belief propagation, not approximated), which is what lets us quantify the recovery as partial rather than merely assert it — but the linear recovery itself is imperfect, and we do not claim the probe reconstructs the posterior exactly. This reproduces the core Simplex belief-geometry phenomenology in a setting where the ground-truth belief state is known exactly.

Methodologically, this paper sits between the two reference points in the bibliography: Cagnetta et al.'s RHM study uses the same hierarchical data model but evaluates the last-token prediction setting, while Shai et al.'s **Transformers learn factored representations** motivates pooling predictive vectors across contexts. Our readout follows the latter all-context-position spirit, treating every prefix position as a belief state to be decoded, while keeping Cagnetta et al.'s exact RHM grammar as the data source.

The most interesting wrinkle is the **inverted strength gradient**: the global root, the spec's nominal target, is the hardest latent to decode, while local near-leaf latents are read off most cleanly. This is intuitive in hindsight — next-token prediction rewards representing whatever is most locally predictive — but it sharpens the belief-geometry claim: the residual stream tracks the **full** latent posterior, weighted toward what the task needs, not a single privileged variable.

Limitations. (i) The grammar is deliberately tiny and fully enumerable ($s = 2$, $L = 3$, 1024 trees); this is what makes the beliefs exact and the verification airtight, but it leaves open how the geometry scales to larger, non-enumerable RHMs. (ii) At $L = 3$ the root posterior collapses to certainty by $k = 3$ on many strings, compressing the dynamic range over which blooming is visible for the root; the spec anticipated this and suggested $L = 4$ as a follow-up. (iii) The strongly-negative mid-layer probe cells indicate the affine probe is locally mis-specified at some position/layer combinations; a per-position or whitened probe would tighten these estimates. (iv) Steering is applied at a single layer (layer 1) along a mean-difference axis; a learned causal direction and a layer sweep would strengthen the causal claim. (v) The detailed pass-1 figures (Figure 2–Figure 6) are from a single training run and grammar draw; the grammar-sweep section quantifies how the **headline** metrics move across 10 grammars and 3 replicate seeds, but the architecture sweep (varying $n_{\text{layer}}, n_{\text{embd}}, n_{\text{head}}$) and larger/longer grammars ($L = 4$) remain future passes.

9. Conclusion

On an exactly-solvable hierarchical grammar, a small transformer's residual stream is a linear image of the exact Bayesian belief simplex: it accumulates across depth, blooms with context, encodes the whole latent hierarchy (local latents most strongly), and is causally used. Pre-registered predictions held in three of four cases, and the one miss — a weaker root probe than predicted — turned out to be a feature of the hierarchy rather than a failure of the belief-geometry hypothesis. The exact, enumerable setting turns a qualitative interpretability story into quantitative, falsifiable measurement.

10. Appendix: Root reconstruction diagnostics

The root-posterior probe is a regression target, but it is also useful to ask the harsher classification question: does the largest coordinate of the linear belief readout recover the sampled root class? Across 400 probe examples and 8 context positions (3200 example-position points), the answer is yes for 1565 points (48.91%). This is well above random guessing, but random guessing is only $1/8 = 12.5\%$ because there are eight root classes; the relevant baseline is not 50%.

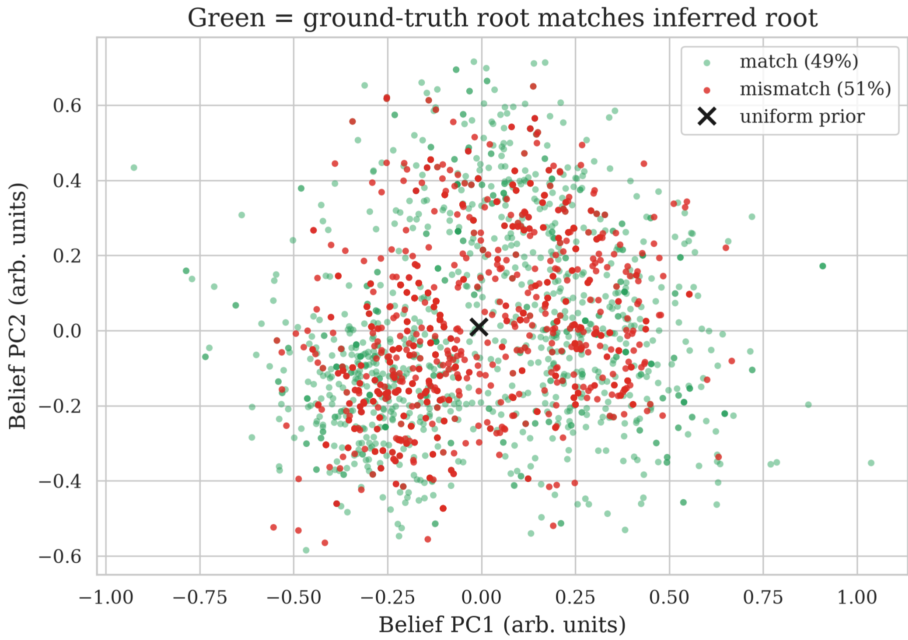

Figure 10: Root reconstruction from the linear belief readout. Green points are example-position pairs where the readout's argmax matches the ground-truth root; red points are mismatches. The overall match rate is 48.91%, compared with a random-guessing baseline of 12.5% (not 50%) for eight root classes.

Restricting to positions where nearly all evidence is visible makes the diagnostic sharper. With seven of eight symbols observed ($k = 6$), the exact Bayesian posterior's MAP root already matches the sampled root in 95.00% of examples, while the linear readout matches in 62.25%. With all eight symbols observed ($k = 7$), the exact posterior is deterministic, but the readout still reaches only 70.50%. Thus early ambiguity explains part, but not all, of the fuzzy root reconstruction.

Subset	Prefix length	Readout match	Exact MAP match
All example-position points	1-8	$\frac{1565}{3200} = 48.91\%$	–
All but last symbol	7	$\frac{249}{400} = 62.25\%$	$\frac{380}{400} = 95.00\%$
Full sequence	8	$\frac{282}{400} = 70.50\%$	$\frac{400}{400} = 100.00\%$

11. Appendix: Per-run grammar-sweep sanity table

Every run in the grammar sweep, no filtering. `grammar` indexes the rule-table draw (`Grammar.random(seed=grammar)`); `seed` is the master replicate seed; the architecture is pinned. `test_ce` is the held-out next-token cross-entropy; `loss_gap_closed` is the fraction of the uniform→Bayes gap closed (sanity covariate, not a gate); `root_r2` and `deepest_r2` are the level-L_0 and mean level-L_2 probe R^2. Loaded directly from `figures/sweep_sanity.csv`.

grammar	seed	n_layer	n_embd	n_head	test_ce	loss_gap_closed	root_r2	deepest_r2
0	0	2	128	4	0.931	0.848	0.242	0.498
0	1	2	128	4	0.891	0.878	0.243	0.497
0	2	2	128	4	0.918	0.858	0.281	0.532
1	0	2	128	4	0.847	0.906	0.375	0.556
1	1	2	128	4	0.812	0.932	0.479	0.616
1	2	2	128	4	0.845	0.908	0.353	0.496
2	0	2	128	4	0.905	0.886	0.402	-0.061
2	1	2	128	4	0.858	0.922	0.454	0.479
2	2	2	128	4	0.866	0.916	0.354	0.46
3	0	2	128	4	0.845	0.91	0.336	0.253
3	1	2	128	4	0.912	0.86	0.333	0.611
3	2	2	128	4	0.92	0.854	0.311	0.607
4	0	2	128	4	0.925	0.876	0.442	0.505
4	1	2	128	4	0.884	0.907	0.416	0.497
4	2	2	128	4	0.909	0.888	0.29	0.442
5	0	2	128	4	0.853	0.911	0.25	0.461
5	1	2	128	4	0.841	0.92	0.277	0.482
5	2	2	128	4	0.848	0.914	0.334	0.493
6	0	2	128	4	0.868	0.896	0.351	0.523
6	1	2	128	4	0.882	0.885	0.4	0.553
6	2	2	128	4	0.836	0.919	0.312	0.523
7	0	2	128	4	0.964	0.82	0.298	0.441
7	1	2	128	4	0.909	0.861	0.256	0.359
7	2	2	128	4	0.892	0.874	0.326	0.436
8	0	2	128	4	0.897	0.863	0.338	0.524
8	1	2	128	4	0.838	0.906	0.359	0.532
8	2	2	128	4	0.868	0.884	0.327	0.532
9	0	2	128	4	0.899	0.894	0.411	0.557
9	1	2	128	4	0.988	0.827	0.394	0.565
9	2	2	128	4	0.87	0.916	0.438	0.604

Table 2: All 30 grammar-sweep runs (10 grammars × 3 seeds), pinned architecture, full 4000-step budget. No convergence filtering.

12. References

Cagnetta, F., Favero, A., Sclocchi, A., and Wyart, M. (2025). **Scaling Laws and Representation Learning in Simple Hierarchical Languages: Transformers vs. Convolutional Architectures**. arXiv:2505.07070.

Shai, A. et al. (2026). **Transformers learn factored representations**. arXiv:2602.02385.

Project sources: `spec.md`, `spec-grammar-sweep.md`, `PREREGISTRATION.md`, `EXECUTION_OUTPUT.md`, `results/analysis.json`, `results/root_reconstruction.json`, `artifacts/train_summary.json`. Grammar sweep: `sweep.py`, `sweep.yaml`, `results/sweep/g*/{config.json,analysis.json}`, `figures/sweep_sanity.csv`.